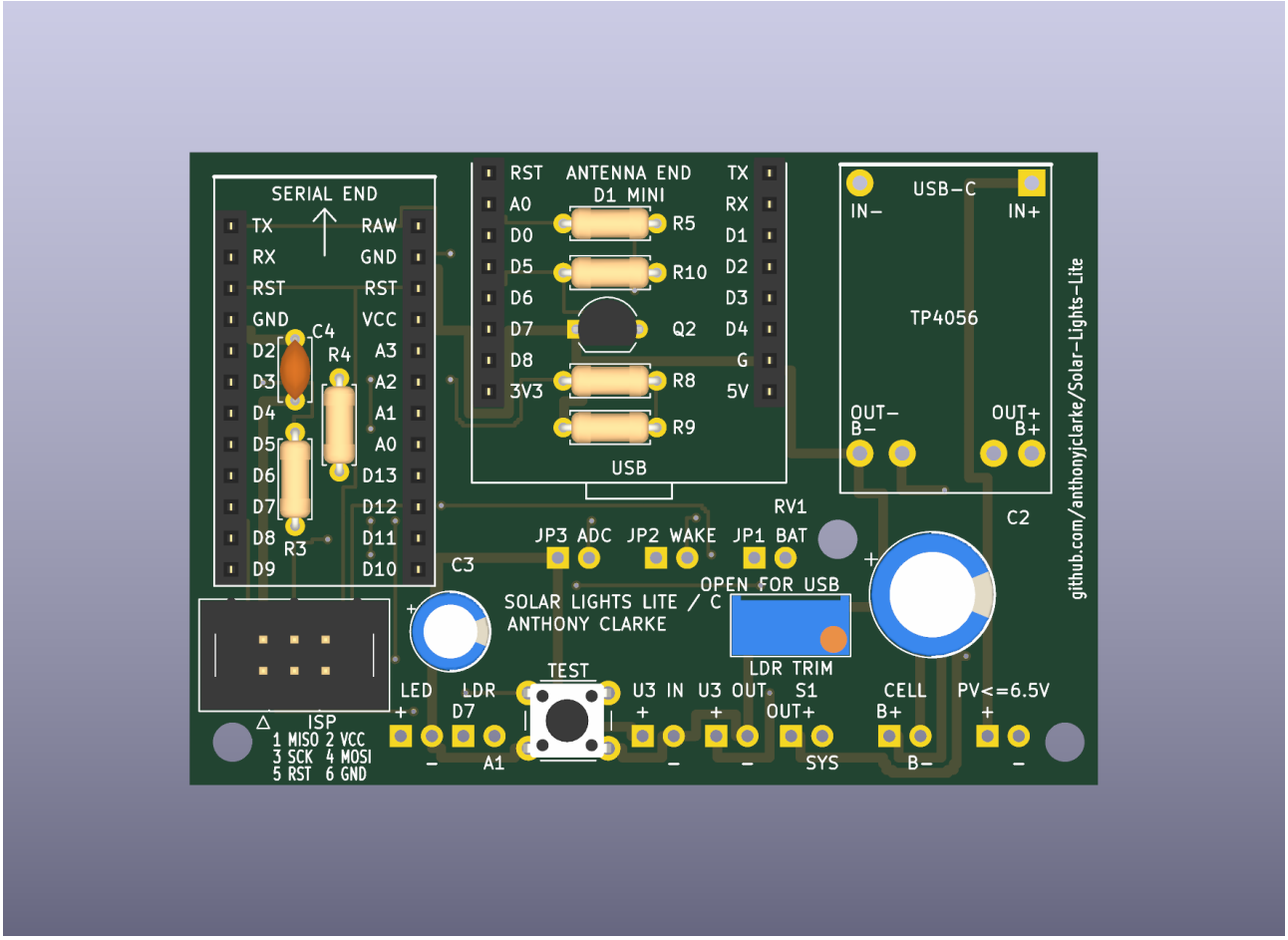
Carrier with charger: layout

28-Oct-2026 | Rev C | CAD checks pass; supplier DFM not run



74 x 51.5 mm carrier (about 3,811 mm2, under half of Rev A). Pro Mini M2 and optional D1 Mini M3 are socketed; the TP4056 charger M1, the TEST button SW1 and the RV1 dusk trimmer sit on the board; every external connection is a 2.54 mm solder point. The image is a KiCad render: modules are not modelled.

External: panel, U0 (5 V buck, if needed), U3 (3.3 V buck-boost), cells, S1, the LED string and the LDR. **No fuses.** B- is its own net and never joins ground; only M1 OUT- is the load return.

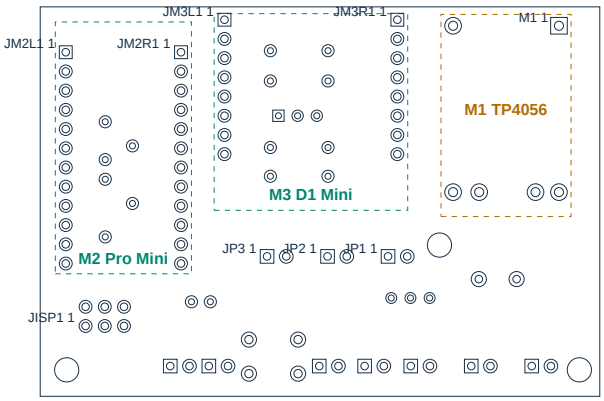
CAD checks: zero ERC/DRC violations, zero unrouted connections and schematic parity issues; 54 independent/model checks recorded.

Before ordering: print review/fit-sheet-A4.pdf at 100% and lay the real modules on it. Pro Mini, D1 Mini and the four TP4056 corner holes were confirmed on the 16-Oct fit sheet; the TP4056 B- and B+ holes were re-measured (3.45 / 7.45 / 3.10 mm) and must now centre on the new sheet. Also confirm 8.5 mm sockets leave about 6 mm under the modules.

Actual-size fit check

28-Oct-2026 | Rev C | CAD checks pass; supplier DFM not run

Print at **Actual size / 100%** (scaling off). Check the 50 mm line, then lay the real modules on the template.



74 x 51.5 mm outline; square pads mark pin 1; amber = TP4056 envelope



50 mm calibration line

M2 Pro Mini: rows 15.24 mm apart, pitch 2.54, body 18 x 33.3 mm, pin 1 about 4.0 mm from the serial end; serial header end up (heavy line). Parts under it: R3, R4, C4.

M3 D1 Mini: rows 22.86 mm apart; body 25.6 x 34.2 mm; antenna end overhangs the top edge, USB end faces the board. Parts under it: R5, R8, R9, R10, Q2.

M1 TP4056: 17.2 x 26.73 mm; holes 14 mm apart at the USB end, rows 22 mm apart, bottom holes 3.45 / 7.45 / 3.10 mm (measured); hole 1.2 mm. See review/fit-sheet-A4.pdf for a 1:1 check.

Parts under a module need under about 6 mm of height with 8.5 mm sockets; C2, C3, RV1 and SW1 stay beside the modules.

ISP lies on its side below the Pro Mini, outlined for an optional DC3-6P box header (key slot down); a plain 2x3 header fits the same holes.

Mounting: 3.2 mm holes at (3.5, 48), (71.3, 48), (52.8, 31.5) mm from the upper-left corner.

Qualification and service

28-Oct-2026 | Rev C | CAD checks pass; supplier DFM not run

Tier	JP1 power	JP2 wake	JP3 ADC
0 standalone	Open	Open	Open
1 USB telemetry	OPEN	Open to flash	Closed while powered
2 battery telemetry	Closed; USB absent	Closed after flashing	Closed

1. Module fit
Measure the TP4056 pad holes first. Verify M2/M3 rows, body outlines and socket height against page 2.

2. Charger prep
Replace the module PROG resistor for 100-150 mA before soldering it on. Meter IN- to OUT- to see whether the module joins them.

3. Unpowered continuity
No rail short; B- not joined to ground; S1 open isolates OUT+ from VBAT_SYS; JP1/JP3 isolate their feeds.

4. In-circuit ISP
Open S1, then program M2 through JISP1. If the programmer supplies VCC it must be set to 3.3 V.

5. LED current
Record peak current at 100% duty: 3.40 V ____ mA; 3.70 V ____ mA; 4.20 V ____ mA; warm ____ mA. Target <= 20 mA.

6. Telemetry and supply
Q2 EBC, inverted 9600-baud reception, ADC calibration; U3 output ____ V; sleep current ____ uA.

7. Charger and outdoor soak
Original Gates 0-8: charge current/termination, cold panel Voc, leakage, temperature limits, energy balance.

Known limits: no fuses (only the module protection), no temperature charge inhibit, no power-path charger, no automatic USB/U3 interlock, no protection against an LED cable short at D9. USB-C on M1 shares the solar input: never use both.

Builder: _____ Date: _____
Fit checked: ____ Electrical gates passed: ____ Ordering release: ____